(a) Flow matching for action generation

training: conditional paths
noise ε ∼ N(0, I) (grey cloud, left)
action distribution (orange, bimodal, right)
straight lines $x_\tau = (1-\tau)\varepsilon + \tau A$

sampling: learned velocity field
grey arrows = $v_\theta(x, \tau \mid o)$
curved ODE trajectories,
Euler dots, N = 10 steps

same process on an action chunk
chunk curves at τ = 0, 0.3, 0.7, 1
noise → smooth trajectory
(x-axis: time step 1…H)

train: $min_\theta \; E_{\tau,\varepsilon,A} \; \| v_\theta(x_\tau, \tau \mid o) - (A - \varepsilon) \|^2$

sample: $x_0 = \varepsilon, \quad x_{\tau+\delta} = x_\tau + \delta \; v_\theta(x_\tau, \tau \mid o), \quad \delta = 1/N$

(b) Distillation: N steps → 1 step

2-D plane, same noise x_0
teacher: curved path, 10 Euler dots
student: one straight jump
to teacher endpoint

$L = \| S_\varphi(x_0, o) - \Phi_T^{(N)}(x_0, o) \|^2$

NFE: N → 1, needs teacher

(c) One-step objective: mean velocity

true path with tangent v (thin)
chord = average velocity
$u(x_0, 0, 1)$ (thick)
one jump: ε → action

$u(x_\tau, \tau, \tau') = (\tau' - \tau)^{-1} \int v \, ds$

$\hat{A} = \varepsilon + u_\theta(\varepsilon, 0, 1 \mid o)$

(d) Warm start & cross-step iteration

loop over control cycles k
shift A_{k-1} → re-noise to τ_0
K = 1–2 denoise steps with o_k
execute s actions → next cycle

$\| T_o(A) - T_o(A') \| \le \rho \| A - A' \|, \; \rho < 1$

holds only for a fixed observation o

o_k changes ⇒ fixed point $A^*(o_k)$ moves